

Theoretical Foundations of Reinforcement Learning:

Week 1

Arya Patil — 24B3943

What is Reinforcement Learning?

Reinforcement Learning (RL) is a framework for sequential decision making where an agent learns by interacting with an environment. Unlike supervised learning, there is no supervisor — only a reward signal. The agent receives observations and rewards, executes actions, and aims to maximise cumulative future reward. Feedback is delayed, and the agent's actions affect subsequent data it receives.

RL is grounded in the **reward hypothesis**: all goals can be described as the maximisation of expected cumulative reward.

Return

The **return** G_t is the total discounted future reward from timestep t :

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

where $\gamma \in [0, 1]$ is the discount factor. When $\gamma = 0$, the agent only cares about immediate reward. When $\gamma \rightarrow 1$, future rewards are weighted nearly equally to present ones. In practice $\gamma \approx 0.99$ is typical.

Policy

A **policy** π is the agent's behaviour function — a mapping from states to actions.

- **Deterministic:** $a = \pi(s)$
- **Stochastic:** $\pi(a \mid s) = P[A = a \mid S = s]$

The goal of RL is to find the optimal policy π^* that maximises expected return from every state.

Value Function

The **state-value function** $V_\pi(s)$ estimates expected return starting from state s under policy π :

$$V_\pi(s) = E_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s \right]$$

The **action-value function** $Q_\pi(s, a)$ estimates expected return of taking action a in state s then following π :

$$Q_\pi(s, a) = E_\pi[G_t \mid S_t = s, A_t = a]$$

Both satisfy the **Bellman expectation equation**:

$$V_\pi(s) = E_\pi[R_{t+1} + \gamma V_\pi(S_{t+1}) \mid S_t = s]$$

Exploration vs Exploitation

A fundamental challenge in RL is balancing exploration and exploitation.

- **Exploitation:** use current knowledge to maximise reward
- **Exploration:** try new actions to discover better strategies, at the cost of short-term reward

A purely exploitative agent may converge to a suboptimal policy. A purely exploratory agent never leverages what it has learned. Common strategies to balance both include ϵ -greedy (explore randomly with probability ϵ , exploit otherwise) and Upper Confidence Bound (UCB), which gives a bonus to less-explored actions.